

Computer Vision

Unit Test-2

Instructions:

- The Question paper contains 5 questions (0-4)
- Each student must **answer only 1 question**.
- The question assigned depends on your **roll number**:

$$\text{Question Number} = (\text{Last Digit of Your Roll Number}) \bmod 5$$

- **Question Structure:**
Each question has two parts:
 - **a) Algorithm Explanation** – Explain the concept or algorithm in 6–7 statements.
Be clear, point-wise and concise.
 - **b-d) Calculation/Problem Solving** – Solve the example or perform the calculation based on the algorithm.
- **Use diagrams or tables** wherever necessary to illustrate your answers.
- **Calculators or computers** are allowed.
- **No collaboration** is allowed.

Question 0:
HOG (Histogram of Oriented Gradients) - Feature Analysis

Given a 4×4 grayscale image patch:

1	2	3	4
13	14	15	16
9	10	11	12
5	6	7	8

- Explain HOG Algorithm: Step-by-Step and its uses in Computer Vision.
- Compute the gradient magnitude and orientation at the **center pixels** 14, 15, 10, 11 using **Sobel operators**:

$$G_x = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

$$G_y = \begin{array}{|c|c|c|} \hline -1 & -2 & -1 \\ \hline 0 & 0 & 0 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

- Using the gradient magnitudes and orientations, create a 4-bin histogram of orientations (0–45°, 45–90° 90–135°, 135–180°) for the center cells.
 - Normalize the histogram obtained using L2 normalization.
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Question 1:
LBP (Local Binary Patterns) - Feature Analysis

Given a 4×4 grayscale image patch:

1	2	3	4
13	14	15	16
9	10	11	12
5	6	7	8

- a. Explain LBP Feature Analysis: Step-by-Step and its uses in Computer Vision.
 - b. Compute the LBP code for the center pixels 14, 15, 10, 11 (use 8 neighbors and threshold by center value).
 - c. Convert each LBP binary code to decimal.
 - d. Build a histogram of these 4 center pixels' LBP values.
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Question 2:
CCA (Connected Component Analysis) - Binary Image Analysis

Given a 4×4 **grayscale image patch**:

15	2	16	12
13	14	1	4
3	5	11	9
10	6	8	7

- a. Explain Connected Component Analysis procedure briefly and its uses in Computer Vision.
 - b. Convert the given grayscale image to binary by thresholding method with threshold = 8
 - c. Count the number of connected components using 4-connectivity and label them.
 - d. Count the number of connected components using 8-connectivity and label them.
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Question 3:
Shape and Region Analysis - Binary Image

Given a 4×4 **binary image patch**:

0	1	1	0
1	1	1	0
0	1	0	0
0	0	0	0

- a. Describe shape and region descriptors, and why are they important in image processing?
 - b. Compute the **basic shape descriptors** of given binary image:
 - i. Area (M_{00}) and Perimeter
 - ii. Sum of x-coordinates, y-coordinates (M_{10} , M_{01})
 - iii. Centroid
 - iv. Second-Order Moments (M_{20} , M_{02} , M_{11})
 - v. Normalized Central Moments (η_{pq})
 - c. Find the **bounding box** of the shape (smallest rectangle that contains all foreground pixels). Compute its width, height, and aspect ratio (width/height).
 - d. Classify the moments (M_{00} , M_{10} , M_{01} , M_{20} , M_{02} , M_{11} , η_{pq}) into different types and its uses.
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Question 4:
Histogram Equalization - Image Enhancement

Given a 8×8 gray image patch with K=8 gray levels:

1	3	0	6	2	3	5	3
4	5	7	0	1	7	3	2
3	6	5	1	5	7	0	3
2	1	3	2	0	4	5	7
1	2	3	4	5	6	2	1
0	7	2	5	2	1	0	7
4	4	1	6	4	1	2	6
6	7	7	6	3	6	5	0

- Explain the Histogram Equalization algorithm briefly, including its purpose in image enhancement.
- Compute the **histogram** of the given image (count of each gray level 0–7).
- Compute the **cumulative distribution function** (CDF) for the histogram.
- Compute the **new pixel values** after histogram equalization using $L=4$:
$$S_k = \text{round}((L-1) \cdot \text{CDF}(k))$$
- Form the **new equalized image** using the transformed pixel values.